DO NOW — RECALL

Scope & parameter passing in C#

Name: _____ Class: _____ Date: _____

1. What is the difference between a C# function and a procedure-style method?

2. Which C# loop always executes its body at least once?

3. What does == do in a C# condition?

4. In Triple(7) , identify the argument. In Triple(int number) , identify the parameter.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

A function returns a value . A procedure-style method uses void and does not return a value.

2. Model answer

do...while Its condition is checked after the body has run.

3. Model answer

It compares two values for equality . = assigns; == compares.

4. Model answer

Argument = 7 · parameter = number The argument is the actual value supplied; the parameter is the named input in the method definition.

By the end, you can…

• explain the scope of local and global/class-level variables;

• explain what happens when a parameter is passed by value;

• use C# ref to pass a variable by reference.

Vocabulary — know these terms

Recall the term — Scope

Recall the term — Local variable

Recall the term — Global variable

Recall the term — Parameter

Recall the term — Argument

Recall the term — Pass by value

Recall the term — Pass by reference

Recall the term — ref

NEW LEARNING

Read, annotate and explain

Scope & parameter passing

```
public class Program
{
public static int score = 0;

public static void Main()
{
int attempts = 3;
}

public static void Change(ref int number)
{
number = 100;
}
}
```

OCR idea: global variables can be accessed across the program; local variables exist only within the subroutine/block in which they are declared.

Scope controls where a variable can be used

A local variable is declared inside a method or block and can only be used within that scope.

For this course, we use a class-level static field to demonstrate the OCR idea of a global variable that can be accessed by multiple methods.

Parameters are local to the method

A parameter receives data when a method is called and behaves as a local variable inside that method.

number exists within Triple . The value supplied in the call is the argument.

By value = the method receives a copy

Changing number changes only the method's local copy. The original variable in Main() is unchanged.

ref gives access to the original variable

Write ref in both places: in the method parameter and in the method call.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Why doesn't the original value change?

Start in Main

```
int score = 10;
```

The variable score belongs to Main() and currently stores 10.

The method receives a copy

```
public static void ChangeByValue(int number)
{
    number = 100;
}
```

The parameter number is a separate local variable initialised with the value 10.

Only number becomes 100

```
ChangeByValue(score);

inside method:
number = 100

back in Main:
score = 10
```

When the method ends, its local parameter is no longer needed.

Reference passing changes the original

```
ChangeByReference(ref score);

// method
public static void ChangeByReference(ref int number)
{
    number = 100;
}
```

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

1. Identify the scope

Which variable can be accessed by multiple methods? Which is local to AddOne() ?

Method / working:

2. Predict pass-by-value

What is output?

Method / working:

3. Make the change persist

Modify the previous example so x becomes 10 after the call.

Method / working:

4. Explain by value vs by reference

OCR expects more than the words “copy” and “address”. Explain what happens to the original data .

Method / working:

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Local variables in two methods
Create two procedure-style methods: First() and Second() .
Inside each method declare a local variable called result , but give it a different value. Output each value.

Question 2

Prove pass-by-value
Set int score = 20; in Main() .
Create SetToZero(int number) that changes number to 0.
Call the method, then output score .

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Change the original with ref
Adapt Challenge 2 so the original score becomes 0.
Rename the method SetToZeroByReference and use ref correctly in both required places.

Question 4

Swap two values
Create Swap(ref int first, ref int second) .
Use one local temporary variable so that the two original values exchange places.
In Main() , start with a = 10 and b = 25 , call the method and output both values.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Passing an integer parameter by value means changing the parameter does not change the original integer variable.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.